15/06/26 Day 20 Kamineni Dhana Sri keerthi

option 02

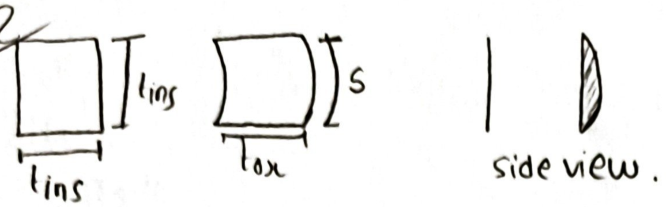


Diagram → refer Day18.

1. Assumming the current is distributed uniformly over cross section of pin hole. [bent rectangular prism] 

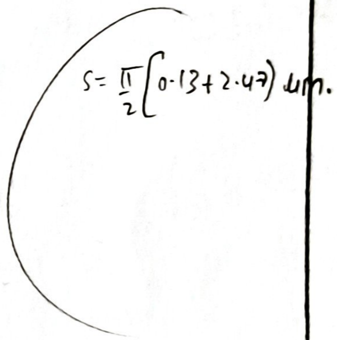
2. L_p → Total height of pinhole [BRP]
   t_{ox} → thickness (radical width)
   S → Arc length
   J → current density.

Inductance of finite thin sheet.

3. Cross sectional Area outer bent rectangular prism

$$A = S L_p \quad (m^2).$$

$$S = \theta [t_{ox} + r_{TSV}] \, (m)$$



4. The total self inductance of a uniform conductor

$$L_{sht} = L_{int} + L_{ext}$$

5. At high frequency  calculating inductance of finite thin sheet.

→ Why we are taking inductance of thin sheet instead of entire 3D mode

$t_{ox} = 0.13\mu m$. $S = 4.08\mu m$. $L_p = 4\mu m$.

Ans:. In semiconductor tech, t_{ox} is usually μ often, but. L_p and S are typically much larger.

Because t_{ox} is so small compared to other two dimensions, the 3D volume behaves electrically almost identical to a 2D sheet.

15/06/

General Lext (sheet/Block finite)

$$L_{ext} = \frac{\mu_0 t}{2\pi}\left[\ln\left[\frac{2(l+w)}{t}\right] + \frac{1}{2}\right] \quad (H)$$

→ Grover's Inductance calculations.

(So when current direction (t) is much smaller than cross section dimensions (l and w) the neumann integral for external flux simplifiers to.)

General Lint

$$L_{int} = \frac{\mu_0 t}{3(l+w)}$$

Now our model

↳ google books.

$$L_{ext} = \frac{\mu_{TSV} t_{ox}}{2\pi}\left[\ln\left[\frac{2[L_p+s]}{t_{ox}}\right] + \frac{1}{2}\right]$$

$$L_{int} = \frac{\mu_{TSV} t_{ox}}{3[L_p+s]}.$$

$$L_{ext} = \frac{\mu_0\, path}{2\pi}\left[\ln\left[\frac{2\, perimeter}{path}\right] + \frac{1}{2}\right]$$

$$l_{xt} = \frac{\mu_0\, path}{3(perimeter)}$$

$$\frac{t_{in}}{\sqrt{t_s^2 + t_{in}^2}} \qquad \frac{0.5\,\mu}{\sqrt{(u)^2\mu + (0.5)^2\mu}}$$

$$L = L_{ext} + L_{int}$$

Proposed model.

$$L_{shl} = \frac{\mu_{TSV} t_{ox}}{2\pi}\left[\ln\left[\frac{2[L_p+S]}{t_{ox}}\right] + \frac{1}{2}\right] + \frac{\mu_{TSV} t_{ox}}{3[L_p+S]}.$$

Novel model

$$L_{shl} = \mu_{TSV} t_{ox}\left[\cosh^{-1}\left[\frac{S}{\sqrt{L_p^2+S}} - \sqrt{t_{ox}^2+L_p^2+s^2} + \frac{S}{4} + \sqrt{L_p^2+s^2}\right]\right].$$

15/06/26 Day 20 Ramineni Dhana sri keerthi

Defected TSV [R_{sht}, L_{sht})

RLGC → same as defect free.

Novel model :- (final)

$$R_{sht} = \frac{\rho_T \cdot t_{iox}}{L_S^2}$$

$$L_{sht} = \frac{\mu_0 \mu_{TSV} t_{ox}}{2\pi}\left[\cosh^{-1}\left(\frac{t_{ins}}{\sqrt{L_S^2 + t_{ins}^2}}\right) - \sqrt{2t_{ins}^2 + L_S^2} + \frac{t_{ins}}{4} + \sqrt{L_S^2 + t_{ins}^2}\right]$$

Proposed model

$$R_{sht} = \frac{\rho_T t_{ox}}{L_P[r_{TSV} + t_{ox}]\theta}$$

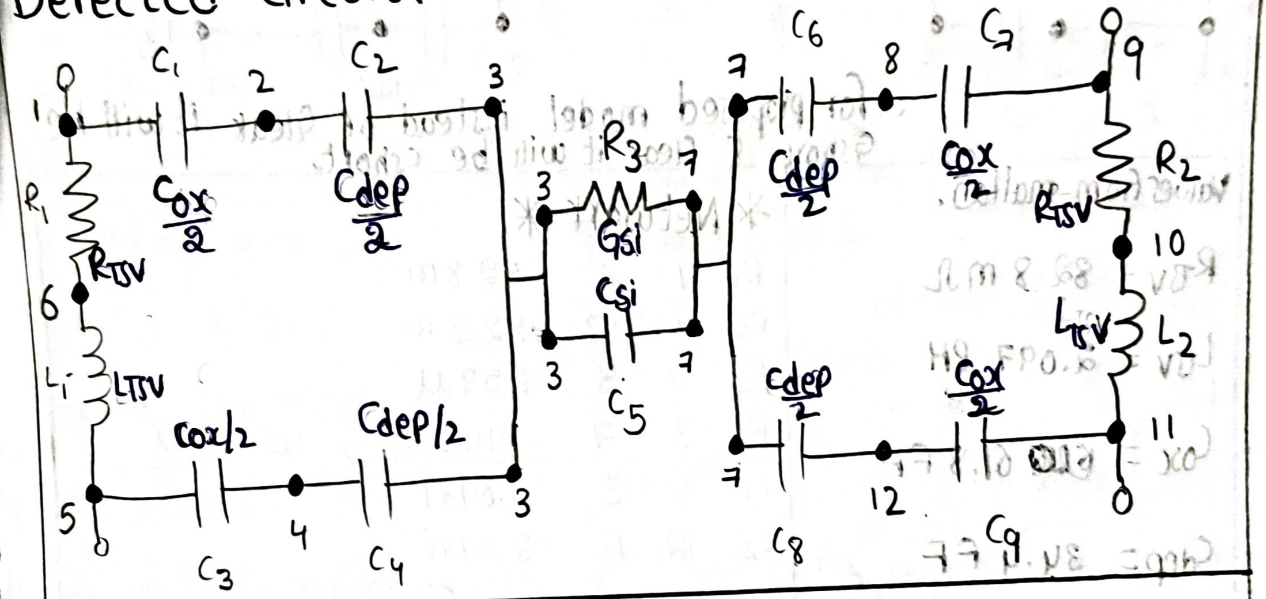
$$L_{sht} = \frac{\mu_0 \mu_{TSV} t_{ox}}{2\pi}\left[\cosh^{-1}\left(\frac{S}{\sqrt{L_P^2 + S^2}}\right) - \sqrt{t_{ox}^2 + L_P^2 + S^2} + \frac{S}{4} + \sqrt{L_P^2 + S^2}\right]$$

6/26, Day 20 Kamineni Ohana Sri keerthi

TSV models

- Defect free
- Defected CG [Chao liu , Proposed]
- Defected RL [Novel paper , proposed]

Defected Circuit.



Values from matlab.

$R_{TSV} = 82.8\,m\Omega$

$L_{TSV} = 2.097\,PH$

$C_{ox} = 65.07\,fF$

$C_{dep} = 35.84\,fF$

$g_{si} = 354\,\mu S$

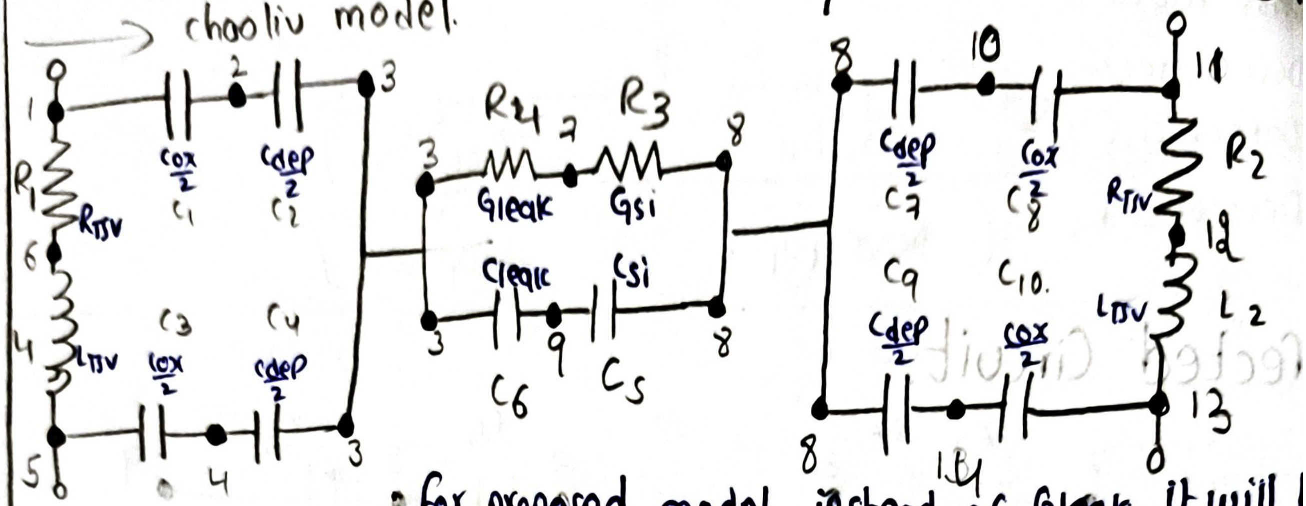
$C_{si} = 3.73\,fF$

Network

R_1	1	6	82.8 m
R_2	9	10	82.8 m
R_3	3	7	2354 u
L_1	6	5	2.097 p
L_2	10	11	2.097 p
C_1	1	2	32.53 f
C_2	2	3	17.92 f
C_3	5	4	32.53 f
C_4	4	3	17.92 f
C_5	3	7	3.73 f
C_6	7	8	17.92 f
C_7	8	9	32.53 f
C_8	7	12	17.92 f
C_9	12	11	32.53 f.

Defected CG. [Chao liu model / Proposed model]

→ chaoliu model.



∴ for proposed model instead of Gleak it will be Gshort & Cleak it will be Cshort.

Values from matlab.

R_{TSV} = 82.8 mΩ

L_{TSV} = 2.097 PH

Cox = ~~[illegible]~~ 61.8 FF 30.9 f

Cdep = 34.4 FF 17.2f

Csi = 2.73 FF

Cleak = 64.0 FF

Gsi = 259 µS ~~[illegible]~~

Gleak = −214.3 µS

Gshort = 108.4 µS

Cshort = 14.09 FF

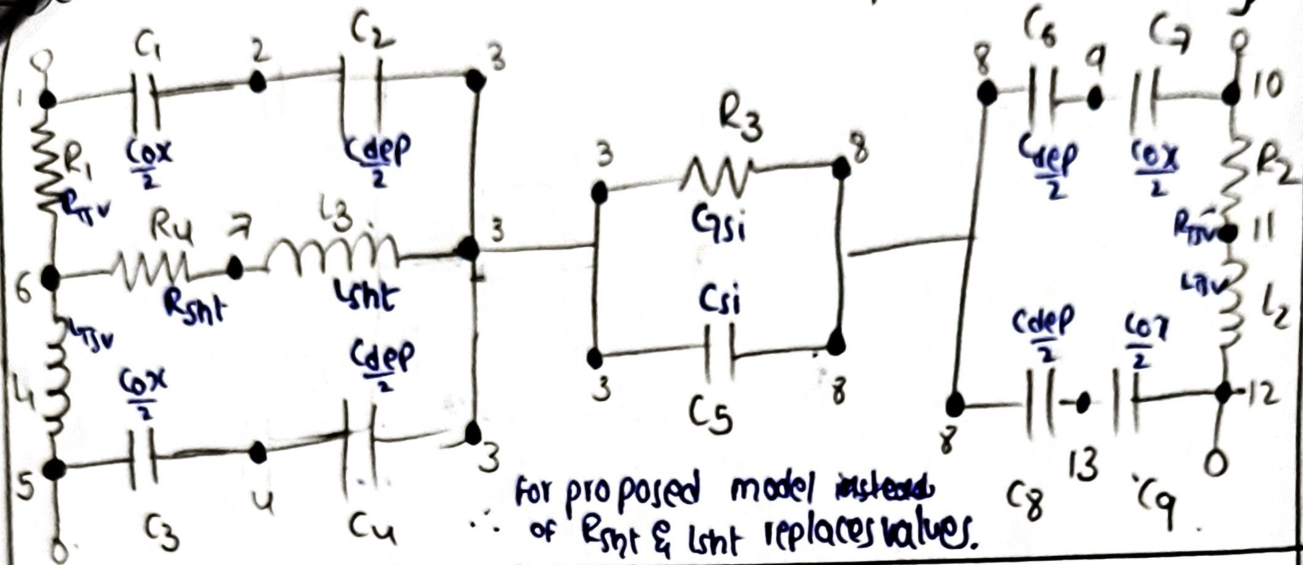
* Network *

R_1	1	6	82.8m
R_2	11	12	82.8 m
R_3	7	8	259 µ
R_4	3	7	−214.3µ / 108.4 µ
L_1	6	5	2.097P
L_2	12	13	2.097P
C_1	1	2	30.9 F
C_2	2	3	17.2F
C_3	5	4	30.9F
C_4	4	3	17.2F
C_5	9	8	2.73f
C_6	3	9	64.0f / 14.09f.
C_7	8	10	17.2f
C_8	10	11	30.9f
C_9	8	14	17.2f
C_{10}	14	13.	30.9f.

proposed

0.007×10^{22} H

efected RL [Novel prosen model / proposed model]



Values from matlab.

$R_{TSV} = 82.8 m\Omega$

$L_{TSV} = 2.097 PH$

$C_{ox} = 65.07 FF$ 32.53f

$C_{dep} = 35.84 FF$ 17.92f

$C_{si} = 3.73 FF$

$G_{si} = 354 \mu\Omega$

→ Novel model

R_{sht} = ~~1.36~~ 0.136 mΩ
0.79 x 10^{-21}

$L_{sht} = 119.011 \times 10^{-21} H$
17

→ Proposed model

$R_{sht\_p} = 0.133 m\Omega$.

$L_{sht\_p} = 20.508 \times 10^{-21} H$.

* Network *

R_1	1	6	82.8m
R_2	10	11	82.8m
R_3	3	8	35uμ
R_4	6	7	0.136m /0.133m.
L_1	6	5	2.097P
L_2	11	12	2.097P
L_3	7	3	1.901e-20 /2.05e-20.
C_1	1	2	32.53f
C_2	2	3	17.92f
C_3	5	4	32.53f
C_4	4	3	17.92f
C_5	3	8	3.73f
C_6	8	9	17.92f
C_7	9	10	32.53f
C_8	8	13	17.92f
C_9	13	12.	32.53F.

...nt have ~~black~~ spikes. what does this mean.

...fect free TSV

3.a → negligible delay. almost no distortion.

3.b → 1. exhibit short +ve & -ve spikes ~~during switching~~ because The TSV capacitance charges & discharges.

2. The +ve spices correspond to charging events
-ve " " " dis " "

3. The similarity btw Ip and Op I ~~..~~ states that TSV. introduces minimal additional losses.

How these currents can reduce.

1. ↓ parasitic Capacitance and inductance.

~~8~~

use ↓ lowdielect.
TSV diameter

F = leakage factor $\frac{\text{damaged area}}{\text{side wall are}}$

~~why~~ rise and ~~fall~~ negative

Rise time ↑ because charging C takes long time

fall time ↓ discharging fast.

1) objective correction

power.

2) 2 TSV (delay / crosstalk / losses) s parameters

performance analysis of bump n TSV

3) parasitics values Table represent per unit.

→ Shivangi Chandrakar

4) during fabrication. identify which process it may occur. which step RL

which step CG

SiV

delay

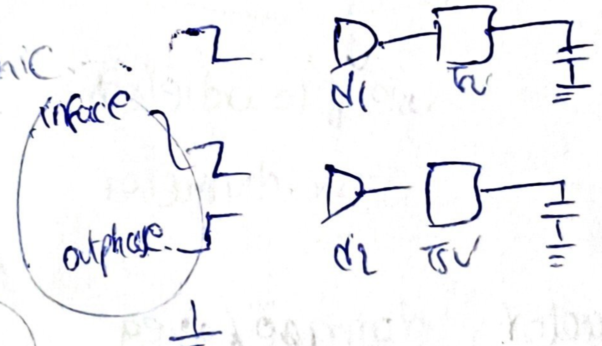
crosstalk. → what is dynamic crosstalk.

delay — +ve / -ve

&

Glitch

dynamic



functional

inp → d1

GN → D2